



General Sir John Kotelawala Defence University

Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 13

1. An online store records details of 5 customer orders:

- Order IDs
- Item Prices (Rs.)
- Quantities

Tasks:

- Declare **three 1D arrays** to store:
 - Order IDs
 - Prices
 - Quantities
- Input all values using a loop
- Declare another array to store **total amount for each order**
- Write a **user-defined function** to calculate total amount for each order
- Call the function and store results
- Display:
 - Order ID
 - Total amount for each order

2. A fitness center tracks activity details of 4 students:

- Student Names
- Steps Walked
- Calories Burned

Tasks:

- Declare **three 1D arrays** to store:
 - Student names
 - Steps

- c. Calories
- ii. Input all values using a loop
- iii. Write a **user-defined function** to analyze the data
- iv. The function should:
 - a. Display students who walked **more than 5000 steps**
 - b. Display students who burned **more than 200 calories**
- v. Call the function and display results

3. A school records marks of 3 students in 4 subjects (Maths, Science, English, ICT).

Tasks:

- Declare a 3×4 2D array
- Input marks using nested loops
- Display marks in matrix form

4. A supermarket tracks sales (in Rs.) of 4 products (Rice, Sugar, Milk, Bread) over 3 days.

Tasks:

- Declare a 3×4 array
- Input daily sales using nested loops
- Display the sales in matrix format

5. A hospital records temperature readings of 3 patients at 4 different times (Morning, Noon, Evening, Night).

Tasks:

- Declare a 3×4 array
- Input temperature readings
- Display all readings in matrix form

6. A bus company records number of passengers traveling on 3 routes over 4 time slots.

Tasks:

- Declare a 3×4 array
- Input number of passengers
- Display the data in matrix form
- Calculate and display **total passengers per route**

7. An electricity board records power usage (units) of **3 houses** over **4 weeks**.

Tasks:

- i. Declare a **3 × 4 2D array** to store usage
- ii. Input data using nested loops
- iii. Display the data in matrix form
- iv. Create functions:
 - a. `int calculateTotal(int arr[3][4], int houseIndex);`
 - b. `float calculateAverage(int total);`
- v. Display:
 - Total usage for each house
 - Average usage per house

8. A university records attendance of **3 students** across **4 lectures** (1 = Present, 0 = Absent).

Tasks:

- i. Declare a **3 × 4 2D array**
- ii. Input attendance using nested loops
- iii. Display the matrix
- iv. Create functions:
 - a. `int calculateAttendance(int arr[3][4], int studentIndex);`
 - b. `void checkEligibility(int total);`
- v. Display:
 - Total attendance per student
 - Whether student is eligible (attendance ≥ 3)